

Q-1. How React's Virtual DOM makes updating the UI faster compared to traditional DOM manipulation.

ANS. React uses a Virtual DOM, which is a lightweight copy of the real DOM. When state or props change, React starts a process called Reconciliation. During reconciliation, React creates a new Virtual DOM and compares it with the previous Virtual DOM using the diffing algorithm. It identifies only the changed elements and updates those parts in the real DOM. Since DOM operations are expensive, updating only the necessary elements makes the UI faster and improves performance.